Scientific Research Methodology
2. About research, the structure of a publication

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References

The Craft of Research, Booth, Colomb, Williams, Bizup, Fitzgerald, Chicago Press
Available online: search for it!

Who needs research?

Research is not just for members of a research laboratory...
Example. A journalist: presenting a topic effectively requires substantial preparation—reading and filtering search results

⋆ Every **professional** needs research skills in the "information age"
⋆ We need to evaluate information and communicate it clearly and precisely

⋆ Identify **reliable**, **high-quality** information

Starting a research project

- focusing on a topic
- finding relevant information
- deciding what to do with that information
- identifying the questions that need to be asked

Planning is necessary

- two aspects: planning the research process and planning the writing process
- we begin with **a question**: the answer is not known *a priori*
- plan written work carefully: anticipate readers' expectations
  - a reporter writes an article to communicate information quickly and capture attention
  - an accountant prepares a report so that reviewers can find all the information they need

Thinking in writing

Why must the results of research be written down?

- spoken statements can be imprecise and influenced by the moment
- are we expressing opinions or verified facts?
- **wrong** or even **dangerous** ideas spread because people accept opinions without asking for **evidence**

A researcher always maintains a degree of **scepticism** about what they read and about their own results!

Remember: the output of research is always a written document, prepared according to established conventions.

What is research?

Collecting information to find the answer to a question that solves a problem.

Problem: finding a part for a car
Research: looking for a car parts supplier

Problem: we need a well-known person's date of birth
Research: carrying out an internet search

Problem: we are curious about a new animal species
Research: searching online for articles in academic journals

People seek answers to such questions every day. Authors make written materials (in print, online, etc.) available so that they can be found when the information is needed.

- a researcher writes with a purpose: to be useful when someone seeks that information
- this serves the community: reliable information presented in a useful way

Why write?

- writing helps us **remember**: information we find but do not record may be lost
- writing helps us **understand**: an idea becomes clearer when we put it into words. New connections, implications, and complications emerge!
- writing helps us **test**: putting ideas on paper often reveals points that need explanation and clarification
- writing helps us **share**: unwritten information can only be passed on directly (teaching, discussion). Written information can reach a wider audience!

Why a formal paper?

- a paper is written **for others to read**
- anticipate critical questions:
  - how were the evidence and arguments evaluated?
  - why is this topic relevant?
  - what other ideas did you consider or discard?
- the community addressed by the research has certain expectations!

Different forms of publication

- research article
- popular science article
- research report
- conference paper, etc.

Each genre has standards and expectations concerning the level of the results and how they are presented.

Understanding roles

What is your role?

- providing new information
- answering a question
- offering a solution to a practical problem

What does the reader want?

- to understand a concept
- to solve a practical problem
- to find information
- to be entertained

Elements of a scientific paper

- Title: a concise description of the results
- Abstract: a summary of the paper
- Introduction: existing literature, motivation, and results obtained
- Main text: materials and methods used, previous results drawn on
- Results: a clear description of the research findings, highlighting their novelty
- Discussion: comparison with previous work and future perspectives
- References